University of Findlay

Educational CVE Analysis Tool (VulnEdu Dashboard)

Deepti Veldanda

Abstract

VulnEdu Dashboard is an interactive web-based dashboard designed to help cybersecurity students and educators understand and analyze Common Vulnerabilities and Exposures (CVE) data.

Project Overview:

- Addresses educational gap by transforming complex CVE data into accessible learning resources
- Combines real-time NVD data with historical trend analysis for comprehensive coverage

Technical Foundation: Python Flask backend, Chart.js visualizations, NVD API integration, hosted on Render platform with UptimeRobot monitoring.

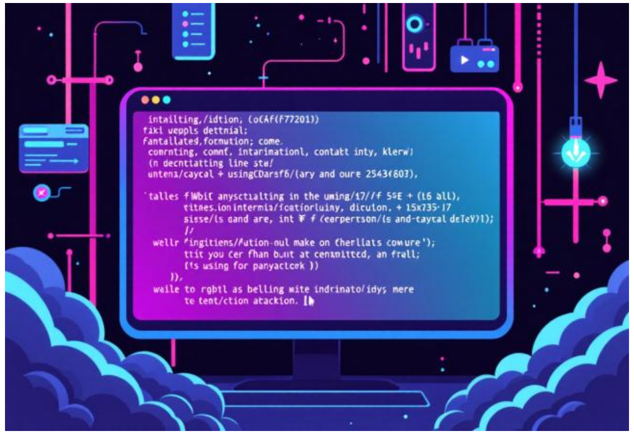
The Problem

Cybersecurity education is complex

Students face steep learning curves when approaching cybersecurity concepts and principles.

Struggling with vulnerability concepts

Abstract technical ideas are difficult to grasp without practical, visual context.



Existing tools are too technical

Current solutions lack accessibility and educational focus for learners.

CVE-2021-44228 Detail

Description

Apache Log4j2 2.0-beta9 through 2.15.0 (excluding security releases 2.12.2, 2.12.3, and 2.3.1) JNDI features used in configuration, log messages, and parameters do not protect against attacker controlled LDAP and other JNDI related endpoints. An attacker who can control log messages or log message parameters can execute arbitrary code loaded from LDAP servers when message lookup substitution is enabled. From log4j 2.15.0, this behavior has been disabled by default. From version 2.16.0 (along with 2.12.2, 2.12.3, and 2.3.1), this functionality has been completely removed. Note that this vulnerability is specific to log4j-core and does not affect log4net, log4cxx, or other Apache Logging Services projects.

Metrics

CVSS Version 4.0 | CVSS Version 3.x | CVSS Version 2.0

NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.

CVSS 3.x Severity and Vector Strings:

NIST: NVD	**Base Score:** 10.0 CRITICAL	**Vector:** CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H
ADP: CISA-ADP	**Base Score:** 10.0 CRITICAL	**Vector:** CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H

QUICK INFO

CVE Dictionary Entry:
CVE-2021-44228
NVD Published Date:
12/10/2021
NVD Last Modified:
08/08/2025
Source:
Apache Software Foundation

Source : https://nvd.nist.gov/vuln/detail/CVE-2021-44228

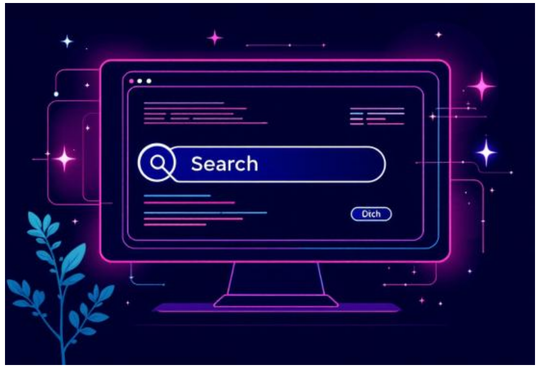
The Solution

Real-time Vulnerability Data

Access the latest security information as it becomes available, keeping you informed of emerging threats immediately.

Interactive Visualizations

Engage with dynamic charts and graphs that make complex vulnerability data easy to understand and analyze visually.



Easy-to-Use Search Functions

Quickly find specific vulnerabilities and related information with intuitive search capabilities, saving time and effort.

The Solution

Educational Learning Modules

Structured lessons guide students through cybersecurity concepts with practical, hands-on examples for effective learning.

Technical Highlights

Technology Stack : Python, Flask, Chart.js, JavaScript, HTML, CSS

Data Foundation

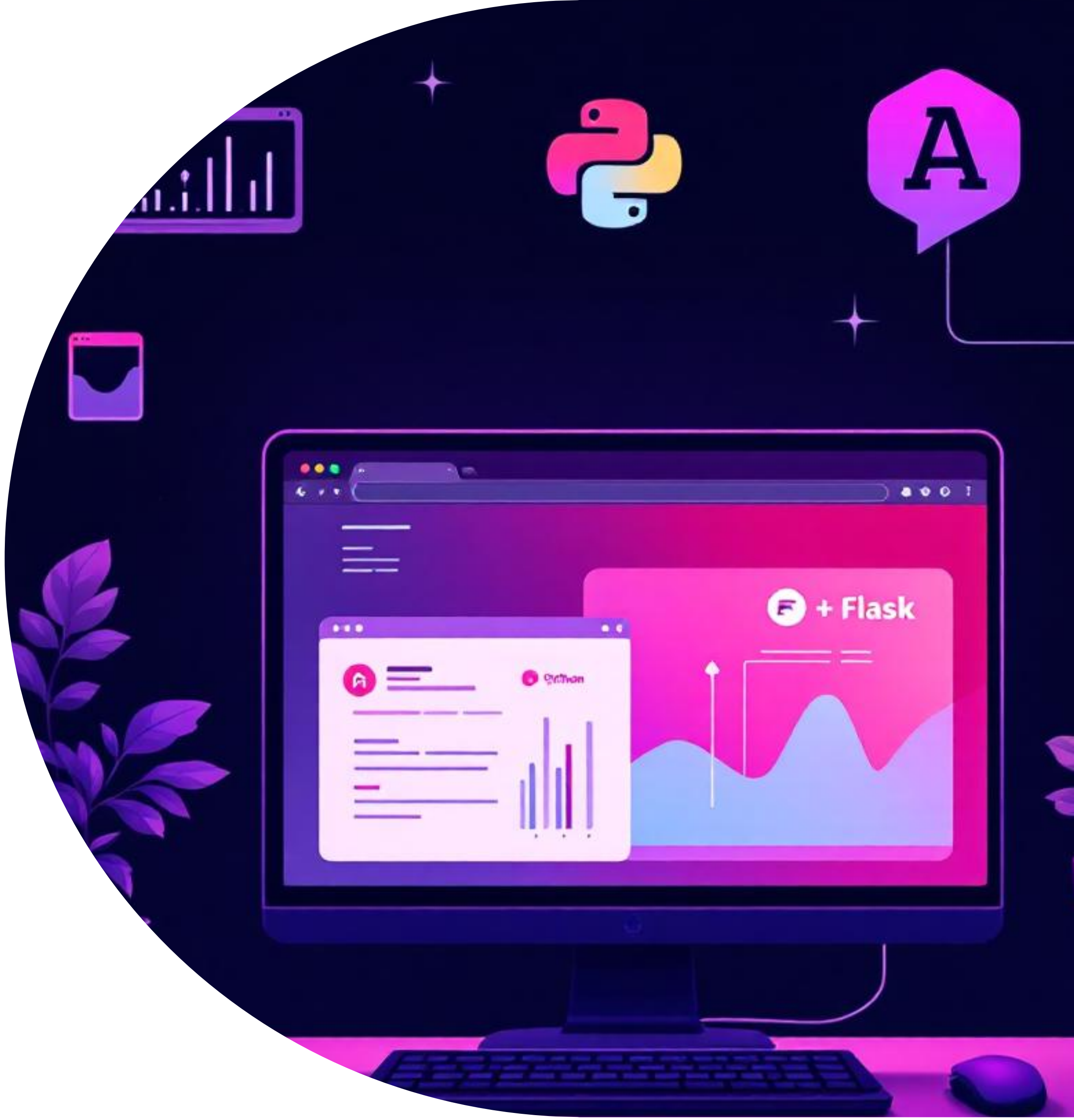
National Vulnerability Database (https://services.nvd.nist.gov/rest/json/cves/2.0)
API Key: Enhanced rate limits (50 requests/30 seconds)

Additional Sources:

CVSS Framework: Severity scoring methodology V3.1

Data Scope:

CVE records from 2010-present
Approximately 150,000+ vulnerability records
CVSS scores, CWE classifications, Severity levels
Publication dates, descriptions, and reference links



Research Questions

What are the most common types of vulnerabilities (CVEs) reported in the last 5 years, and how do they vary by severity?

How do different vulnerability types compare in terms of frequency and severity distribution?

How can an interactive dashboard improve the understanding and prioritization of vulnerabilities for cybersecurity students?

What are the key features of an interactive dashboard that can effectively educate students about CVE trends and patterns?

Beneficiaries

Primary Users:

- **Cybersecurity Students:** Learning vulnerability patterns and security concepts
- **Educators:** Teaching vulnerability analysis and cybersecurity principles
- **Entry-level Security Professionals:** Understanding threat prioritization

Secondary Users:

- **Researchers:** Analyzing vulnerability trends for academic purposes

Impact and Learning

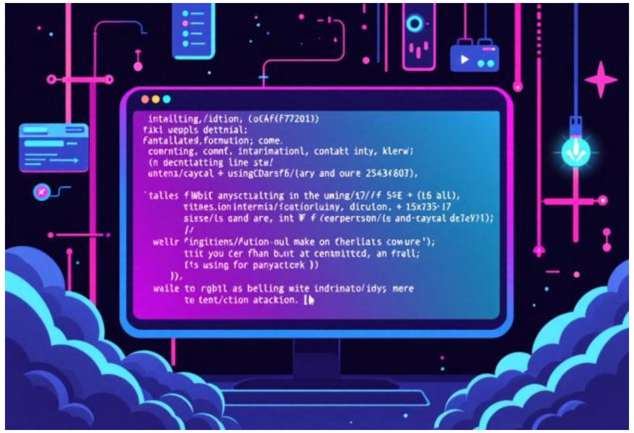
What students gain

Understanding vulnerability patterns

Students learn to recognize common security weaknesses and how they manifest across different systems and applications.

Threat prioritization skills

Develop the ability to assess and rank vulnerabilities by severity, enabling effective resource allocation for security efforts.



Practical cybersecurity knowledge

Gain hands-on experience with real vulnerability data and industry-standard tools, preparing students for professional cybersecurity roles.

Next steps

Short-term Improvements:

- Mobile optimization and user authentication
- Educational gamification with quizzes and badges

Medium-term Goals:

- Machine learning integration and multi-language support
- Enterprise features and open-source development

Long-term Vision:

- Advanced analytics and educational innovation
- Interactive learning platform with comprehensive quiz systems

References

Primary Data Sources:
National Institute of Standards and Technology (NIST) https://nvd.nist.gov/
MITRE Corporation, Common Vulnerabilities and Exposures (CVE). https://cve.mitre.org/
MITRE Corporation, Common Weakness Enumeration (CWE). https://cwe.mitre.org/
FIRST.org, Common Vulnerability Scoring System (CVSS). https://www.first.org/cvss/

Technical Documentation:
Flask Documentation, Pallets Projects https://flask.palletsprojects.com/
Chart.js Documentation. https://www.chartjs.org/docs/
NVD API 2.0 Documentation. https://nvd.nist.gov/developers/vulnerabilities
Render Platform Documentation. https://render.com/docs

Industry Standards:
OWASP Top 10 Web Application Security Risks. https://owasp.org/www-project-top-ten/
SANS Top 25 Most Dangerous Software Errors. https://www.sans.org/top25-software-errors/

Appendix - Technical Specifications

Development Stack:

- **Backend Framework:** Python 3.9+, Flask 2.3.0
- **Data Processing:** Pandas, Requests, JSON parsing libraries
- **Frontend Technologies:** HTML5, CSS3, JavaScript ES6
- **Visualization Library:** Chart.js 3.9 with custom configurations
- **Database:** File-based caching with JSON storage

Deployment Infrastructure:

- **Hosting Platform:** Render.com with automatic deployments
- **Monitoring Service:** UptimeRobot for 24/7 website health monitoring
- **Version Control:** GitHub
- **Configuration Management:** .yml files for deployment and environment settings

VulnEdu

https://vulnedu.onrender.com

Feedback

Help Shape the Future of VulnEdu, Drop an email with your suggestions, feature requests, feedback or ideas to improve the dashboard.

Your input is crucial in making this tool more effective for cybersecurity learning!

Email : Veldandad@findlay.edu